

MODBUS over TCP Support

MODBUS/TCP uses the Ethernet media in physical layers to carry the MODBUS message handling structure and can support a large number of devices in one network; it is easier to integrate into the Local Area Network (LAN) of a company, so it is the choice of more and more customers.

Here, it is used for remote 3rd party monitoring and control. MODBUS TCP is agnostic of the server connection. It works only over LAN. When configured, MODBUS TCP does not initiate a connection. The server waits for a client to connect. Only one connection is supported.



NOTE

The MODBUS TCP function is disabled by default. When enabled, it supports TCP port 502 by default. The port number can be reconfigured.

MODBUS over TCP Configuration

→ To setup MODBUS TCP:

- Select Communication → LAN Conf → Modbus TCP (the default port is 502).

To modify the TCP port, select Modbus TCP → TCP Port, set the port number and long-press Enter.



NOTE

The default device ID of the inverter connected to the Ethernet is 1.

When the MODBUS TCP feature is enabled, the following information is displayed:

■ Status:

- Init – Initializing server – This state only occurs after the first configuration until it reaches the ready status. This activity lasts about 10sec.
- Ready – The server is up and waiting for a client to connect.
- Connected – The client is connected.
- Failed – The server is unable to accept clients (see error message).

■ Error messages:

- Disconnected – The Ethernet cable is not connected
- Gateway Ping Failed – A ping to the 1st router failed
- No IP – Either no DHCP configuration or static IP configuration (no DHCP server that assigned an IP address) or need to define a static IP.



NOTE

The TCP server idle time is 2 minutes. In order to leave the connection open, the request should be made within 2 minutes. The connection can remain open without any MODBUS requests.

Register Mapping – Monitoring Data

This section describes the registers mapping for the inverter monitoring data (read-only MODBUS protocol data). The SolarEdge inverter mapping for monitoring data is based on the open protocol managed by SunSpec: SunSpec Alliance Interoperability Specification – Inverter Models v1.0. Refer to the SunSpec Alliance Interoperability Specification – Common Models (Elements) document for a detailed description of the protocol.

The register mapping can be downloaded from the SunSpec Alliance web page: <http://www.sunspec.org/>.

SolarEdge inverters support the following mappings:

- SunSpec module ID 101, 102¹ and 103 register mappings.
- SolarEdge three phase inverters with Synergy technology also support SunSpec module ID 160 register mappings.

Common Model MODBUS Register Mappings

The base Register Common Block is set to 40001 (MODBUS PLC address [base 1]) or 40000 (MODBUS Protocol Address [base 0]).

All parameters are defined as in the SunSpec Common block definition, except for the **C_Options** register, which is set to NOT_IMPLEMENTED.

- **C_Manufacturer** is set to SolarEdge.
- **C_Model** is set to the appropriate inverter model, e.g. SE5000.
- **C_Version** contains the CPU software version with leading zeroes, e.g. 0002.0611.
- **C_SerialNumber** contains the inverter serial number.
- **C_DeviceAddress** is the device MODBUS ID.

Address		Size	Name	Type	Description
(base 0)	(base 1)				
40000	40001	2	C_SunSpec_ID	uint32	Value = "SunS" (0x53756e53). Uniquely identifies this as a SunSpec MODBUS Map
40002	40003	1	C_SunSpec_DID	uint16	Value = 0x0001. Uniquely identifies this as a SunSpec Common Model Block
40003	40004	1	C_SunSpec_Length	uint16	65 = Length of block in 16-bit registers
40004	40005	16	C_Manufacturer	String(32)	Value Registered with SunSpec = "SolarEdge "
40020	40021	16	C_Model	String(32)	SolarEdge Specific Value
40044	40045	8	C_Version	String(16)	SolarEdge Specific Value
40052	40053	16	C_SerialNumber	String(32)	SolarEdge Unique Value
40068	40069	1	C_DeviceAddress	uint16	MODBUS Unit ID

¹ Supported only in split-phase configurations (Japanese grid and 240V grid in North America)

Inverter Device Status Values

The following I_Status_* values are supported:

Parameter	Value	Description
I_STATUS_OFF	1	Off
I_STATUS_SLEEPING	2	Sleeping (auto-shutdown) – Night mode
I_STATUS_STARTING	3	Grid Monitoring/wake-up
I_STATUS_MPPT	4	Inverter is ON and producing power
I_STATUS_THROTTLED	5	Production (curtailed)
I_STATUS_SHUTTING_DOWN	6	Shutting down
I_STATUS_FAULT	7	Fault
I_STATUS_STANDBY	8	Maintenance/setup

Inverter Model MODBUS Register Mappings

The following table lists the supported MODBUS register values.

Unsupported values are indicated by the NOT_IMPLEMENTED value.

The base register of the Device Specific block is set to 40070 (MODBUS PLC address [base 1]), or 40069 (MODBUS Protocol Address [base 0]).

- **acc32** is a uint32 accumulator that should always increase. Its value is in the range of 0...4294967295.
- **Scale Factors.** As an alternative to floating point format, values are represented by Integer values with a signed scale factor applied. The scale factor explicitly shifts the decimal point to left (negative value) or to the right (positive value).

For example, a value "Value" may have an associated value "Value_SF"

Value = "Value" * 10^{Value_SF} for example:

- For "Value" = 2071 and "Value_SF" = -2 Value = 2071*10⁻² = 20.71
- For "Value" = 2071 and "Value_SF" = 2 Value = 2071*10² = 207100

Address		Size	Name	Type	Units	Description
(base 0)	(base 1)					
40069	40070	1	C_SunSpec_DID	uint16		101 = single phase 102 = split phase 103 = three phase
40070	40071	1	C_SunSpec_Length	uint16	Registers	50 = Length of model block
40071	40072	1	I_AC_Current	uint16	Amps	AC Total Current value
40072	40073	1	I_AC_CurrentA	uint16	Amps	AC Phase A Current value
40073	40074	1	I_AC_CurrentB	uint16	Amps	AC Phase B Current value
40074	40075	1	I_AC_CurrentC	uint16	Amps	AC Phase C Current value
40075	40076	1	I_AC_Current_SF	int16		AC Current scale factor
40076	40077	1	I_AC_VoltageAB	uint16	Volts	AC Voltage Phase AB value
40077	40078	1	I_AC_VoltageBC	uint16	Volts	AC Voltage Phase BC value
40078	40079	1	I_AC_VoltageCA	uint16	Volts	AC Voltage Phase CA value

Address		Size	Name	Type	Units	Description
(base 0)	(base 1)					
40079	40080	1	I_AC_VoltageAN ¹	uint16	Volts	AC Voltage Phase A to N value
40080	40081	1	I_AC_VoltageBN ¹	uint16	Volts	AC Voltage Phase B to N value
40081	40082	1	I_AC_VoltageCN ¹	uint16	Volts	AC Voltage Phase C to N value
40082	40083	1	I_AC_Voltage_SF	int16		AC Voltage scale factor
40083	40084	1	I_AC_Power	int16	Watts	AC Power value
40084	40085	1	I_AC_Power_SF	int16		AC Power scale factor
40085	40086	1	I_AC_Frequency	uint16	Hertz	AC Frequency value
40086	40087	1	I_AC_Frequency_SF	int16		Scale factor
40087	40088	1	I_AC_VA	int16	VA	Apparent Power
40088	40089	1	I_AC_VA_SF	int16		Scale factor
40089	40090	1	I_AC_VAR	int16	VAR	Reactive Power
40090	40091	1	I_AC_VAR_SF	int16		Scale factor
40091	40092	1	I_AC_PF	int16	%	Power Factor
40092	40093	1	I_AC_PF_SF	int16		Scale factor
40093	40094	2	I_AC_Energy_WH	acc32	WattHours	AC Lifetime Energy production
40095	40096	1	I_AC_Energy_WH_SF	uint16		Scale factor
40096	40097	1	I_DC_Current	uint16	Amps	DC Current value
40097	40098	1	I_DC_Current_SF	int16		Scale factor
40098	40099	1	I_DC_Voltage	uint16	Volts	DC Voltage value
40099	40100	1	I_DC_Voltage_SF	int16		Scale factor
40100	40101	1	I_DC_Power	int16	Watts	DC Power value
40101	40102	1	I_DC_Power_SF	int16		Scale factor
40103	40104	1	I_Temp_Sink	int16	Degrees C	Heat Sink Temperature
40106	40107	1	I_Temp_SF	int16		Scale factor
40107	40108	1	I_Status	uint16		Operating State
40108	40109	1	I_Status_Vendor	uint16		Vendor-defined operating state and error codes. For error description, meaning and troubleshooting, refer to the SolarEdge Installation Guide.

¹ Supported only in split-phase configurations (Japanese grid and 240V grid in North America).